Automatica 192 (2026) 113160

Contents lists available at ScienceDirect

Automatica

journal homepage: www.elsevier.com/locate/automatica

An update-resilient Kalman filtering approach ☆

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ARTICLE INFO

Article history:
Received 10 April 2025
Received in revised form 11 January 2026
Accepted 5 May 2026

Keywords:
Robust Kalman filtering
Kullback–Leibler divergence
Minimax game

ABSTRACT

We propose a new robust filtering paradigm considering the situation in which model uncertainty, described through an ambiguity set, is present only in the observations. We derive the corresponding robust estimator, referred to as update-resilient Kalman filter, which appears to be novel compared to existing minimax game-based filtering approaches. Moreover, we characterize the corresponding least favorable model and analyze the filter stability. Finally, some numerical examples show the effectiveness of the proposed estimator.

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1. Introduction

Various state estimation paradigms have been proposed in the literature to tackle model uncertainty such as considering parametric approaches (El Ghaoui & Calafiore, 2001; Kim et al., 2021; Rocha & Terra, 2021), filters robust to outliers (Huang et al., 2016, 2019) and minimax approaches (Hansen & Sargent, 2008; Levy & Nikoukhah, 2013; Nguyen et al., 2023; Yoon et al., 2004). In the latter, uncertainty is characterized by an ambiguity set that captures the "mismatch" between the actual and nominal models. In particular, to allow the uncertainty to be uniformly distributed over time, a well-established paradigm is to define this set at each time step which is a ball whose radius represents the level of uncertainty and its center corresponds to the nominal model (Levy & Nikoukhah, 2013). The minimax paradigm is formulated as a dynamic game: the maximizer selects the least favorable model within the prescribed ambiguity set, while the other player, i.e. the estimator, seeks to minimize the estimation error with respect to the chosen maximizer. Various extensions of this paradigm have been proposed, including ambiguity sets defined using different metrics (Xu et al., 2025; Zorzi, 2016), formulations with degenerate densities (Yi & Zorzi, 2020, 2021), adaptive formulations (Yi et al., 2024) and the case of nonlinear state space models (Longhini et al., 2021). Finally, the relaxation of such minimax games lead to the so called risk sensitive filters (Hassibi et al., 1999; Huang et al., 2018; Speyer & Chung, 2008).

These minimax game-based filters share a common feature: the uncertainties are captured by an ambiguity set defined over the entire state space model, meaning that uncertainties affect both the state dynamics and the observations. This particular feature has a significant impact on the structure of the resulting robust estimators. The latter exhibit a structure similar to that of the Kalman filter (KF), in which the "resilience" to uncertainty is in the prediction stage, that is, the covariance matrix of the prediction error is modified. An exception, however, is represented by Abadeh et al. (2018), where the "resilience" is applied in the update stage. The minimax formulation of this robust filtering problem is based on an ambiguity set defined through the Wasserstein distance. Nevertheless, that paper does not provide an explicit interpretation of the underlying uncertainty in terms of the state space model. Furthermore, it lacks the characterization of the state space representation corresponding to the least favorable scenario, and does not guarantee the filter stability even when the tolerance (i.e. the Wasserstein radius) is small.

In many real-world systems, the state dynamics are relatively well-understood and accurately modeled, while the uncertainty is mainly in the observations. An example is represented by the displacement estimation problem of a mass spring damper system, where the displacement is measured by sensors that are susceptible to various uncertainties. In such unbalanced uncertainty scenarios, where the "dirty" sensor data constitutes the dominant source of uncertainty, it becomes important to reconsider whether characterizing the ambiguity set in terms of the entire state space model remains the most appropriate choice. Indeed, this characterization may still appear reasonable from an intuitive standpoint, since in practice there exists a mismatch (even though mild) between the actual and nominal state process models. Thus, an important question concerns whether the robust filters like (Levy & Nikoukhah, 2013; Xu et al., 2025), which are resilient in the prediction stage, can perform effectively in the presence of unbalanced uncertainty.

☆ The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Andrea Garulli under the direction of Editor Alessandro Chiuso.

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https://doi.org/10.1016/j.automatica.2026.113160
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In this paper, we propose a new robust state space filtering problem in which uncertainties arise in the observation model. To this end, we define a novel ambiguity set that pertains solely to the density characterizing the observation model by means of the Kullback–Leibler (KL) divergence. Our analysis shows that the resulting estimator exhibits a Kalman filter-like structure, in which the resilience is in the update stage. This paradigm is fundamentally different from the ones in Levy and Nikoukhah (2013), Xu et al. (2025). This fact is also reflected in the corresponding least favorable model, whose approximation through a state-space representation exhibits a larger dimension than those in Levy and Nikoukhah (2013), Xu et al. (2025). Moreover, we analyze the filter stability in the worst case scenario. The numerical results show that even in the presence of a mild mismatch between the actual and nominal state process models, the proposed estimator still outperforms those in Levy and Nikoukhah (2013), Xu et al. (2025). This is because in such unbalanced uncertainty scenarios, characterizing the ambiguity set in terms of the entire state space model result in overly conservative estimates, as modelers has opportunity to allocate the "mismatch" budget uniformly across both the state process and the observation model. Furthermore, numerical results show that the proposed estimator performs similarly to the one in Abadeh et al. (2018). Indeed, since both are resilient in the update stage, our theoretical framework suggests that the estimator proposed in Abadeh et al. (2018) implicitly postulates uncertainty only in the observation model. In addition, the experimental evidence indicates that our approach is significantly more computationally efficient than (Abadeh et al., 2018), i.e. a property that is particularly important for real-time applications. Finally, we consider the relaxed version of the aforementioned minimax game and show that the resulting estimator is a new risk sensitive filter.

The outline of the paper is as follows. In Section 2 we introduce the problem formulation. In Section 3 we derive the update-resilient Kalman filter and its corresponding least favorable model. In Section 4 we analyze the stability of the proposed filter. In Section 5 we provide some numerical examples. In Section 6 we derive the corresponding risk sensitive estimation paradigm. Finally, in Section 7 we draw the conclusions.

Notation: Given a symmetric matrix $K : K > 0$ ($K \geq 0$) means that K is positive (semi-)definite; $\sigma_{max}(K)$ is the maximum eigenvalue of K; $\mathrm{tr}(K)$ and $\det(K)$ denote the trace and determinant of K, respectively. Then, $v \sim \mathcal{N}(m, R)$ means that x is a Gaussian random vector with mean m and covariance matrix R. Finally, throughout the paper, we omit the arguments of probability density functions, e.g. ψ_t instead of $\psi_t(y_t|x_t)$, when the context is clear in order to keep the notation concise. Moreover, $\hat{x}_{t|t}$ denotes the posterior state estimate, while $\hat{x}_t$ denotes the prior state estimate. The corresponding covariance matrices follow the same subscript notation: the symbol used for the covariance with subscript t refers to the prior covariance matrix, and the same symbol with subscript $t|t$ refers to the posterior covariance matrix.

2. Problem formulation

We consider a robust discrete-time state space filtering problem where perturbations occur solely in the observations. In this scenario, the actual model for the state process is assumed to be known and takes the form:

$$x_{t+1} = Ax_t + \epsilon_t \tag{1}$$

where $A \in \mathbb{R}^{n\times n}$, $x_t \in \mathbb{R}^n$ is the state vector and $\epsilon_t \in \mathbb{R}^n$ is white Gaussian noise with the zero mean and covariance matrix $Q \in \mathbb{R}^{n\times n}$. Thus, model (1) can be entirely characterized by the following transition density $p_t(x_{t+1}|x_t) \sim \mathcal{N}(Ax_t, Q)$ and the initial state $x_0 \sim p_0(x_0)$, which is assumed to be independent from ϵ_t. The nominal model for the observations is

$$y_t = Cx_t + \varepsilon_t \tag{2}$$

where $C \in \mathbb{R}^{m\times n}$, $y_t \in \mathbb{R}^m$ is the observation vector, and $\varepsilon_t \in \mathbb{R}^m$ is white Gaussian noise with zero mean and covariance matrix $R \in \mathbb{R}^{m\times m}$, which is independent from ϵ_t. We also assume that $Q > 0$ and $R > 0$. The nominal model (2) is completely characterized by the nominal density $\psi_t(y_t|x_t) \sim \mathcal{N}(Cx_t, R)$. Accordingly, the state space model (1)–(2) over the finite time interval $t \in \{0 \dots N\}$ is characterized by the following nominal joint probability density of $X_N := \{x_0 \dots x_{N+1}\}$ and $Y_N := \{y_0 \dots y_N\}$:

$$p(X_N, Y_N) = p_0(x_0) \prod_{t=0}^{N} p_t(x_{t+1}|x_t)\, \psi_t(y_t|x_t).$$

Let $\tilde{\psi}_t(y_t|x_t)$ denote the conditional density characterizing the actual model for the observations. Note that, $\tilde{\psi}_t$ is not required to follow any specific form. We assume that actual density of X_N, Y_N, say $\tilde{p}(X_N, Y_N)$, follows a Markov structure similar to the nominal one:

$$\tilde{p}(X_N, Y_N) = p_0(x_0) \prod_{t=0}^{N} p_t(x_{t+1}|x_t)\, \tilde{\psi}_t(y_t|x_t). \tag{3}$$

We measure the modeling mismatch between the nominal density of X_N, Y_N and the actual one through the KL divergence (Kullback & Leibler, 1951):

$$\mathcal{D}_{KL}(\tilde{p}, p) = \int \tilde{p} \ln\left(\frac{\tilde{p}}{p}\right) dX_N dY_N. \tag{4}$$

Taking the expectation of

$$\ln\left(\frac{\tilde{p}}{p}\right) = \sum_{t=0}^{N} \ln\left(\frac{\tilde{\psi}_t(y_t \mid x_t)}{\psi_t(y_t \mid x_t)}\right)$$

with respect to $\tilde{p}(X_N, Y_N)$, we see the KL divergence (4) takes the form: $\mathcal{D}_{KL}(\tilde{p}, p) = \sum_{t=0}^{N} \mathcal{D}_{KL}(\tilde{\psi}_t, \psi_t)$ where

$$\begin{aligned}\mathcal{D}_{KL}(\tilde{\psi}_t, \psi_t) &:= \tilde{\mathbb{E}}\left[\ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right)\right] \\ &= \iint \tilde{\psi}_t(y_t|x_t) p_t(x_t) \ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right) dy_t dx_t,\end{aligned} \tag{5}$$

and $p_t(x_t)$ denotes the marginal density of x_t. At time t, given the observations $Y_{t-1} := \{y_s,\ s \leq t-1\}$, we assume that the actual density $\tilde{\psi}_t(y_t|x_t)$ belongs to the following convex ambiguity set[1]:

$$\mathcal{B}_t := \left\{ \tilde{\psi}_t \text{ s.t. } \mathcal{D}(\tilde{\psi}_t, \psi_t) \leq c_t \right\} \tag{6}$$

which can be regarded as a "ball", with tolerance $c_t > 0$ representing its radius about the nominal density ψ_t. Such ball is with respect to the metric induced by the KL divergence (5) conditioned on the observations up to time t, i.e.

$$\begin{aligned}\mathcal{D}(\tilde{\psi}_t, \psi_t) &:= \tilde{\mathbb{E}}\left[\ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right) \middle| Y_{t-1}\right] \\ &= \iint \tilde{\psi}_t(y_t|x_t) \tilde{p}_t(x_t|Y_{t-1}) \ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right) dy_t dx_t\end{aligned}$$

where $\tilde{p}_t(x_t|Y_{t-1})$ is the actual *a priori* conditional density of x_t conditioned on Y_{t-1}. The latter is, in general, different from

[1] The adjective ambiguity is used to express the lack of the precise knowledge about the actual model.

the nominal one, say $p_t(x_t|Y_{t-1})$, because the conditioning depends on the model for the observations which is affected by uncertainty. In practice, the tolerance is unknown. One possible strategy is to learn a constant tolerance from data, i.e. $c_t = c$. More precisely, given a dataset of observations and a finite set of candidate values for c, the tolerance is estimated as the candidate corresponding to the robust output predictor that minimizes the sample variance of the output prediction error, Yi et al. (2025). Notice that, the ambiguity set (6) differs from the one considered in Levy and Nikoukhah (2013). While we consider uncertainty only in the measurement Eq. (2), Levy and Nikoukhah (2013) considers uncertainty in both the state process and measurement equations.

The unbalanced uncertainty scenario described above arises in many practical problems. Indeed, the state process model is often assumed to be known, since it is typically derived from well-established physical laws. By contrast, although uncertainty in the observation model can be reduced through the use of advanced sensors, such devices are often very expensive. As a result, less advanced sensors are typically employed in practice due to budgetary constraints. Moreover, sensor uncertainty cannot be entirely eliminated, even with advanced hardware, because of unexpected faults, such as mechanical stress or calibration drift, or environmental factors, such as multipath effects in GPS and reflections in LiDAR.

The significance of the ambiguity set. Consider (2) as a regression model where x_t acts as the input of such regression model, whose prior is defined by (1), and y_t is the output. Note that, the latter is completely described by ψ_t. Assume to collect a set of M independent state-output data corresponding to time t, i.e. $\mathbf{D}_{t,M} := \{(y_t^k, x_t^k),\ k = 1 \dots M\}$. We assume these data are generated by the actual model with $Y_{t-1} = \{y_0 \dots y_{t-1}\}$ fixed. The log-likelihood of the data based on the regression model (2) is $\ell(\mathbf{D}_{t,M}; \psi_t) := \sum_{k=1}^{M} \ln \psi_t(y_t^k|x_t^k)$. Thus, the expected log-likelihood is

$$\begin{aligned}\ell_\infty(\psi_t) &:= \lim_{M\to\infty} \frac{1}{M}\ell(\mathbf{D}_{t,M}; \psi_t) = \tilde{\mathbb{E}}\left[\ln \psi_t|Y_{t-1}\right] \\ &= \iint \ln(\psi_t)\, \tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})dy_t dx_t \\ &= -\iint \tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1}) \ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right) dy_t dx_t \\ &\quad + \iint \ln(\tilde{\psi}_t)\tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})dy_t dx_t \\ &= \kappa_t - \mathcal{D}(\tilde{\psi}_t, \psi_t)\end{aligned}$$

where the limit above almost surely exists (i.e. equality holds with probability one) and

$$\kappa_t := \iint \ln(\tilde{\psi}_t)\tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})dy_t dx_t$$

is a term not depending on the regression model (2).

We conclude that the ambiguity set (6) includes models whose expected log-likelihood is bounded below by a threshold determined by the tolerance c_t:

$$\ell_\infty(\tilde{\psi}_t) \geq \kappa_t - c_t.$$

It is worth noting that the ambiguity set in (6) is fundamentally different from the one proposed in Levy and Nikoukhah (2013), Xu et al. (2025), i.e.

$$\breve{\mathcal{B}}_t := \left\{(\psi_t, \tilde{p}_t) \text{ s.t. } \iiint \tilde{\psi}_t\tilde{p}_t \ln\left(\frac{\tilde{\psi}_t\tilde{p}_t}{\psi_t p_t}\right) dy_t dx_t dx_{t+1}\right\}, \tag{7}$$

which considers uncertainty in both $\psi_t(y_t|x_t)$ and $p_t(x_{t+1}|x_t)$. However, the latter is an inappropriate choice when the primary source of uncertainty lies in ψ_t, i.e. the measurement equation. In such cases, designing a minimax estimator that allows uncertainty in both $\psi_t(y_t|x_t)$ and $p_t(x_{t+1}|x_t)$ grants the hostile player the freedom to allocate a relevant portion of the mismatch budget to the state dynamics (1), which misrepresents the actual scenario. As a result, the performance of the estimator will be severely compromised.

The aim of this paper is to address the following problem.

Problem 1. Design a robust estimator of x_t given Y_t that minimizes the upper bound on the variance of the filtering error for all models whose conditional density of the observation model belongs to the ambiguity set (6).

In what follows, to ease the exposition, we will consider the case in which the state space model (1)–(2) is time-invariant, however the results we will present can be straightforwardly extended to time-varying case.

3. Update–resilient Kalman filter

We design the robust estimator of x_t given Y_t as the solution to the following dynamic minimax game:

$$(\tilde{\psi}_t^\star, g_t^\star) = \arg\min_{g_t\in\mathcal{G}_t} \max_{\tilde{\psi}_t\in\mathcal{B}_t} J_t(\tilde{\psi}_t, g_t) \tag{8}$$

where the objective function is given by

$$\begin{aligned}J_t(\tilde{\psi}_t, g_t) &= \frac{1}{2}\tilde{\mathbb{E}}\left[\|x_t - g_t(y_t)\|^2 |Y_{t-1}\right] \\ &= \frac{1}{2}\iint \|x_t - g_t(y_t)\|^2\, \tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})\, dy_t dx_t\end{aligned}$$

which represents the variance of the filtering error under the actual density $\tilde{\psi}_t$. The latter is restricted to lie within the ambiguity set $\mathcal{B}_t$, i.e. we impose an upper bound on the mismatch between $\tilde{\psi}_t$ and the nominal density ψ_t which is measured using the conditional KL divergence $\mathcal{D}(\tilde{\psi}_t, \psi_t)$. $\mathcal{G}_t$ denotes the class of estimators with finite second-order moments with respect to all the densities $\tilde{\psi}_t\tilde{p}_t(x_t|Y_{t-1})$ such that $\tilde{\psi}_t \in \mathcal{B}_t$. Note that, $\tilde{\psi}_t$ must satisfy also the normalization constraint:

$$I_t(\tilde{\psi}_t) := \iint \tilde{\psi}_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})\, dy_t dx_t = 1. \tag{9}$$

Note that, J_t is quadratic in g_t, in particular it is convex in g_t, and linear in $\tilde{\psi}_t$. Then, on the basis of the Von Neumann's minimax theorem (Aubin & Ekeland, 2006), there exists a saddle point $(\tilde{\psi}_t^\star,\ g_t^\star)$ such that:

$$J_t(\tilde{\psi}_t,\ g_t^\star) \leq J_t(\tilde{\psi}_t^\star,\ g_t^\star) \leq J_t(\tilde{\psi}_t^\star,\ g_t) \tag{10}$$

for any $g_t \in \mathcal{G}_t$ and $\tilde{\psi}_t \in \mathcal{B}_t$. From the second inequality of (10), it follows that the minimizer of (8) coincides with the expectation of x_t taken with respect to the actual filtering density, i.e. $\tilde{p}_t(x_t|Y_t)$, which depends on $\tilde{\psi}_t^\star$. Next, we characterize the solution to the minimax problem (8).

Lemma 2. *Under the assumption that $\tilde{p}(x_t|Y_{t-1})$ is different from zero almost everywhere, the maximizer of (8) takes the form:*

$$\tilde{\psi}_t^0(y_t|x_t) = \frac{1}{M_t(\lambda_t)} \exp\left(\frac{1}{2\lambda_t}\|x_t - g_t(y_t)\|^2\right) \psi_t(y_t|x_t) \tag{11}$$

where

$$M_t(\lambda_t) = \iint \exp\left(\frac{1}{2\lambda_t}\|x_t - g_t(y_t)\|^2\right)\psi_t\tilde{p}_t(x_t|Y_{t-1})\, dy_t dx_t \tag{12}$$

is the normalizing constant such that (9) holds. Moreover, $\lambda_t > 0$ is the unique solution to the equation $\mathcal{D}(\tilde{\psi}_t^0, \psi_t) = c_t$.

Proof. We want to maximize $J_t(\tilde{\psi}_t, g_t)$ with respect to $\tilde{\psi}_t \in \mathcal{B}_t$ using the Lagrangian multipliers theory. More precisely, we consider the Lagrange function:

$$\begin{aligned}&\mathcal{L}(\tilde{\psi}_t,\ \lambda_t,\ \beta_t)\\&= J_t(\tilde{\psi}_t, g_t) + \lambda_t(c_t - \mathcal{D}(\tilde{\psi}_t, \psi_t)) + \beta_t(I_t(\tilde{\psi}_t) - 1)\\&= \iint \left(\frac{1}{2}\|x_t - g_t(y_t)\|^2 - \lambda_t \ln\left(\frac{\tilde{\psi}_t}{\psi_t}\right) + \beta_t\right)\\&\quad \times \tilde{\psi}_t\tilde{p}_t(x_t|Y_{t-1})\,dy_t dx_t + \lambda_t c_t - \beta_t\end{aligned} \tag{13}$$

where $\lambda_t > 0$ is the Lagrange multiplier corresponding to constraint $\tilde{\psi}_t \in \mathcal{B}_t$ and β_t is the one corresponding to constraint (9). Notice that, the Lagrangian (13) is concave with respect to $\tilde{\psi}_t$ because it is the sum of two linear functions in $\tilde{\psi}_t$, i.e. J_t and I_t, and $-\mathcal{D}(\tilde{\psi}_t, \psi_t)$ which is concave with respect to the second argument, Cover and Thomas (1991). We show that $\mathcal{L}_t$ has a unique stationary point, then the latter is the unique point of maximum for $\mathcal{L}_t$. The first variation of $\mathcal{L}_t$ along the direction $\delta\tilde{\psi}_t$ is given by

$$\begin{aligned}\delta\mathcal{L}(\tilde{\psi}_t, \lambda_t, \beta_t; \delta\tilde{\psi}_t) &= \iint \delta\tilde{\psi}_t\tilde{p}_t(x_t|Y_{t-1})\\&\times \Big(\frac{1}{2}\|x_t - g_t(y_t)\|^2 + \lambda_t \ln\left(\frac{\psi_t}{\tilde{\psi}_t}\right) + \beta_t - \lambda_t\Big)dy_t dx_t.\end{aligned}$$

Accordingly, the stationary point $\tilde{\psi}_t^0$ must satisfy $\delta\mathcal{L}(\tilde{\psi}_t^0, \lambda_t, \beta_t; \delta\tilde{\psi}_t) = 0$ for any function $\delta\tilde{\psi}_t$. Since $\tilde{p}(x_t|Y_{t-1})$ is different from zero almost everywhere, we obtain

$$\frac{1}{2}\|x_t - g_t(y_t)\|^2 + \lambda_t \ln\left(\frac{\psi_t}{\tilde{\psi}_t^0}\right) + \beta_t - \lambda_t = 0.$$

Then, it is easy to see that

$$\tilde{\psi}_t^0(y_t|x_t) = \psi_t \exp\left(\frac{1}{2\lambda_t}\|x_t - g_t(y_t)\|^2 + \frac{\beta_t}{\lambda_t} - 1\right).$$

It is not difficult to see that the optimal value for β_t, say β_t^0, is such that $\exp(\beta_t^0/\lambda_t - 1) = M_t(\lambda_t)^{-1}$ where $M_t(\lambda_t)$ is defined in (12). Thus, we obtain (11).

It remains to consider the dual problem for the Lagrange multiplier $\lambda_t > 0$. More precisely, the dual function is given by:

$$\begin{aligned}\tilde{\mathcal{L}}(\lambda_t) &= \mathcal{L}(\tilde{\psi}_t^0, \lambda_t, \beta_t^0)\\&= \iint \left(\frac{1}{2}\|x_t - g_t(y_t)\|^2 - \lambda_t \ln\left(\frac{\tilde{\psi}_t^0}{\psi_t}\right)\right)\\&\qquad \times \tilde{\psi}_t\tilde{p}_t(x_t|Y_{t-1})\,dy_t dx_t + \lambda_t c_t\\&= \lambda_t(\ln(M_t(\lambda_t)) + c_t).\end{aligned}$$

Notice that, $\exp\left(\frac{1}{2\lambda_t}\|x_t - g_t(y_t)\|^2\right) \to 1$ as $\lambda_t \to \infty$. Thus, $\ln(M_t(\lambda_t)) \to 0$, so $\tilde{\mathcal{L}}(\infty) = \infty$ since $c_t > 0$. Accordingly, the infimum cannot be attained for $\lambda_t \to \infty$. Finally, we consider:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\lambda_t}\tilde{\mathcal{L}}(\lambda_t) &= \ln(M_t(\lambda_t)) + \lambda_t M_t^{-1}(\lambda_t)\frac{\mathrm{d}}{\mathrm{d}\lambda_t}M_t(\lambda_t) + c_t\\&= c_t - \lambda_t^{-1}J_t(\tilde{\psi}_t^0(\lambda_t), g_t) + \ln(M_t(\lambda_t))\\&= c_t - \mathcal{D}(\tilde{\psi}_t^0(\lambda_t), \psi_t).\end{aligned}$$

In the case $\tilde{p}_t(x_t|Y_{t-1})$ is Gaussian and g_t is an affine function of y_t (and these conditions are both satisfied in the proof of Theorem 3 to characterize the saddle point), then it is not difficult to see that for c_t taken sufficiently small, there exists $\bar{\lambda}_t > 0$ such that $\frac{\mathrm{d}}{\mathrm{d}\lambda_t}\tilde{\mathcal{L}}(\lambda_t) < 0$ for $\lambda_t = \bar{\lambda}_t$. Accordingly, the infimum cannot be attained for $\lambda_t \le \bar{\lambda}_t$. Since $\tilde{\mathcal{L}}(\lambda_t)$ is a continuous function for $\lambda_t > \bar{\lambda}_t$, by Weierstrass' theorem it follows that $\tilde{\mathcal{L}}$ admits a point of minimum in $(\bar{\lambda}_t, \infty)$. Moreover, it is not difficult to see that $\tilde{\mathcal{L}}$ is strictly convex and thus the point of minimum is the unique stationary point. Imposing the stationarity condition we immediately see that λ_t must satisfy condition $\mathcal{D}(\tilde{\psi}_t^0, \psi_t) = c_t$. □

It is worth noting that $\tilde{\psi}_t^\star$ and $g_t^\star$ are mutually dependent. In particular, the minimax problem (8) could have more than one saddle point solution. In what follows, we derive the minimizer of (8), for $t \in \{0 \dots N\}$, corresponding to the initial state x_0 which is Gaussian distributed. More precisely, the idea is to substitute the structure of the maximizer $\tilde{\psi}_t^0$, as defined in Lemma 2, into Problem (8), obtaining an alternative formulation in which the minimizer remains unchanged. This alternative formulation allows us to characterize the minimizer.

Theorem 3. *Consider the estimation problem corresponding to the state space model (1)–(2) and whose update estimate is obtained through (8). Assume that:*

$$p_0(x_0) \sim \mathcal{N}\left(\hat{x}_0, P_0\right). \tag{14}$$

Then, the estimator of x_t given Y_t is

$$\hat{x}_{t|t} = \hat{x}_t + L_t(y_t - C\hat{x}_t), \tag{15}$$

where L_t is the filtering gain:

$$L_t = P_t C^\top (CP_t C^\top + R)^{-1}. \tag{16}$$

Moreover, the posterior error covariance matrix corresponding to the least favorable posterior density is given by

$$V_{t|t} = (P_{t|t}^{-1} - \lambda_t^{-1} I)^{-1} \tag{17}$$

where

$$P_{t|t} = (I - L_t C)P_t \tag{18}$$

denotes the posterior error covariance matrix corresponding to the nominal posterior density and $0 < \lambda_t < \sigma_{max}(P_{t|t})$ is the unique solution to

$$\frac{1}{2}\left(\ln\det(I - \lambda_t^{-1}P_{t|t}) + \operatorname{tr}\left((I - \lambda_t^{-1}P_{t|t})^{-1} - I\right)\right) = c_t. \tag{19}$$

Moreover,

$$\tilde{p}_t(x_{t+1}|Y_t) \sim \mathcal{N}\left(\hat{x}_{t+1}, P_{t+1}\right), \tag{20}$$

the predictor of x_{t+1} given Y_t takes the form

$$\hat{x}_{t+1} = A\hat{x}_{t|t}, \tag{21}$$

and its corresponding prediction error covariance matrix is

$$P_{t+1} = AV_{t|t}A^\top + Q. \tag{22}$$

Proof. We prove (20) using the induction principle. Condition (20) with $t = -1$ holds by (14) since there are no observations for conditioning. Next, we prove that if $\tilde{p}_t(x_t|Y_{t-1}) \sim \mathcal{N}\left(\hat{x}_t, P_t\right)$ for $t \ge 0$, then $\tilde{p}_{t+1}(x_{t+1}|Y_t)$ will be Gaussian. Let $w_t := \begin{bmatrix} x_t^\top & y_t^\top \end{bmatrix}^\top$. In accordance with the model for the observations (2), the nominal density of w_t given Y_{t-1} is

$$p_t(w_t|Y_{t-1}) = \psi_t(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1}) \sim \mathcal{N}(m_t, K_t)$$

with

$$m_t = \begin{bmatrix} m_{x_t} \\ m_{y_t} \end{bmatrix} = \begin{bmatrix} \hat{x}_t \\ C\hat{x}_t \end{bmatrix},$$

$$K_t = \begin{bmatrix} K_{x_t} & K_{x_t y_t} \\ K_{y_t x_t} & K_{y_t} \end{bmatrix} = \begin{bmatrix} P_t & P_t C^\top \\ CP_t & CP_t C^\top + R \end{bmatrix}.$$

Then, in view of Lemma 2, the least favorable density of w_t given Y_{t-1} takes the following form

$$\begin{aligned}\tilde{p}_t(w_t|Y_{t-1}) &= \tilde{\psi}_t^0(y_t|x_t)\tilde{p}_t\,(x_t|Y_{t-1}) \\ &= \frac{1}{M_t(\lambda_t)}\exp\left(\frac{1}{2\lambda_t}\,\|x_t - g_t(y_t)\,\|^2\right) p_t\,(w_t|Y_{t-1})\,. \end{aligned} \tag{23}$$

Since $p_t(w_t|Y_{t-1})$ is Gaussian, in view of (23), it follows that $\tilde{p}_t(w_t|Y_{t-1})$ is Gaussian. Hence, we have

$$\begin{aligned}&\mathcal{D}(\tilde{p}_t(w_t|Y_{t-1}),\ p_t(w_t|Y_{t-1})) \\ &\quad := \iint \tilde{p}_t(w_t|Y_{t-1})\ln\left(\frac{\tilde{p}_t(w_t|Y_{t-1})}{p_t(w_t|Y_{t-1})}\right)dw_t \\ &\quad = \iint \tilde{\psi}_t^0(y_t|x_t)\tilde{p}_t(x_t|Y_{t-1})\ln\left(\frac{\tilde{\psi}_t^0}{\psi_t}\right)dy_t dx_t \\ &\quad = \mathcal{D}(\tilde{\psi}^0,\,\psi_t).\end{aligned}$$

Such identity determines the ambiguity set for $\tilde{p}_t(w_t|Y_{t-1})$ corresponding to $\mathcal{B}_t$:

$$\tilde{\mathcal{B}}_t = \{\tilde{p}_t(w_t|Y_{t-1}),\ s.t.\ \ \mathcal{D}(\tilde{p}_t(w_t|Y_{t-1}), p_t(w_t|Y_{t-1})) \le c_t\}.$$

Thus, by plugging $\tilde{\psi}_t^0$ into problem (8), we obtain the alternative minimax formulation:

$$\min_{g_t\in\mathcal{G}_t}\ \max_{\tilde{p}_t(w_t|Y_{t-1})\in\tilde{\mathcal{B}}_t}\ \tilde{J}_t(\tilde{p}_t(w_t|Y_{t-1}), g_t) \tag{24}$$

where the corresponding objective function is given by:

$$\tilde{J}_t = \frac{1}{2}\int \|x_t - g_t\,(y_t)\|^2\,\tilde{p}_t(w_t|Y_{t-1})dw_t. \tag{25}$$

In (24) the minimizer remains the same as in (8), while the equivalent maximizer is $\tilde{p}_t(w_t \mid Y_{t-1}) \in \tilde{\mathcal{B}}_t$. Since both $p_t(w_t|Y_{t-1})$ and $\tilde{p}_t(w_t|Y_{t-1})$ are Gaussian, then by Levy and Nikoukhah (2004, Theorem 1) we obtain that the maximizer of (24), hereafter called $\tilde{p}_t^0(w_t|Y_{t-1})$, takes the form

$$\tilde{p}_t^0(w_t|Y_{t-1}) \sim \mathcal{N}(\tilde{m}_t, \tilde{K}_t) \tag{26}$$

with

$$\tilde{m}_t = m_t = \begin{bmatrix} m_{x_t} \\ m_{y_t}\end{bmatrix},\quad \tilde{K}_t = \begin{bmatrix} \tilde{K}_{x_t} & K_{x_t y_t} \\ K_{y_t x_t} & K_{y_t}\end{bmatrix}.$$

Moreover,

$$\tilde{K}_{x_t} = V_{t|t} + L_t K_{y_t x_t} L_t^\top$$

where $L_t = K_{x_t y_t}K_{y_t}^{-1}$, $V_{t|t} = (P_{t|t}^{-1} - \lambda_t^{-1}I)^{-1}$. Notice that, the original covariance matrix $P_{t|t}$ (corresponding to the nominal posterior density) is modified by incorporating λ_t^{-1} into $V_{t|t}$. The variable λ_t denotes the Lagrange multiplier corresponding to the constraint $\tilde{p}_t(w_t|Y_{t-1}) \in \tilde{\mathcal{B}}_t$ (and corresponding to the constraint $\tilde{\psi}_t \in \mathcal{B}_t$ in the original problem), whose optimal value is determined by solving (19); the latter also characterizes the least favorable density in (26). Since the model for the state process is not affected by uncertainties, we have that:

$$\tilde{p}_t(x_{t+1}, w_t|Y_{t-1}) = \tilde{p}_t^0(w_t|Y_{t-1})p_t(x_{t+1}|x_t)$$

which is Gaussian. Thus, the corresponding marginal densities $\tilde{p}(x_{t+1}, y_t|Y_{t-1})$ and $\tilde{p}(y_t|Y_{t-1})$ are both Gaussian. Then, we obtain that

$$\tilde{p}_t(x_{t+1}|Y_t) = \tilde{p}_t(x_{t+1}, y_t|Y_{t-1})/\tilde{p}_t(y_t|Y_{t-1})$$

is also Gaussian. Accordingly, we conclude that (20) holds. Since the maximizer (26) is Gaussian, then the corresponding minimizer of (24), takes the form in (15)–(16). Then, (21) and (22) follow from the fact that $\tilde{p}_t(x_{t+1}|Y_t)$ is Gaussian. □

Algorithm 1 U-RKF at time step t

Input $\hat{x}_t$, P_t, c_t, y_t, A, Q, C, R
Output $\hat{x}_{t|t}$, $\hat{x}_{t+1}$

1: $L_t = P_t C^\top (CP_tC^\top + R)^{-1}$
2: $\hat{x}_{t|t} = \hat{x}_t + L_t(y_t - C\hat{x}_t)$
3: $P_{t|t} = P_t - P_tC^\top(CP_tC^\top + R)^{-1}CP_t$
4: Find θ_t s.t. $\gamma(P_{t|t}, \theta_t) = c_t$
5: $V_{t|t} = (P_{t|t}^{-1} - \theta_t I)^{-1}$
6: $\hat{x}_{t+1} = A\hat{x}_{t|t}$
7: $P_{t+1} = AV_{t|t}A^\top + Q$

Theorem 3 essentially adopts the least favorable density within the ambiguity set characterized in (6), which ultimately leads to the least favorable posterior error covariance in (17). The resulting estimator is outlined in Algorithm 1 where $\theta_t := \lambda_t^{-1}$ is called risk sensitivity parameter and

$$\gamma(P,\theta) := \frac{1}{2}\left(\ln\det(I-\theta P) + \operatorname{tr}\left((I-\theta P)^{-1} - I\right)\right). \tag{27}$$

Observe that by setting $\lambda = \lambda_t^{-1}$ and $P = P_{t|t}$ in (27), we obtain the left-hand side of (19). The proposed approach differs from the classical Kalman filter in the update stage, specifically in Steps 4 and 5 of Algorithm 1, which correspond to a portion of that stage. More precisely, the risk-sensitivity parameter θ_t is incorporated into the filtering error covariance $V_{t|t}$. This modification effectively changes the filtering gain and makes the update stage more conservative when facing model uncertainties. As θ_t increases, the eigenvalues of the filtering covariance matrix $V_{t|t}$ becomes more inflated, enabling the filter to hedge against potential modeling errors. In plain words, the risk-sensitivity parameter determines how cautious the filter becomes in the presence of model uncertainty. Since this robustness mechanism is embedded solely in the update stage, we refer to our filtering approach as update-resilient Kalman filter (U-RKF). As we already pointed out in Section 2, there exists a fundamental difference between the proposed ambiguity set and the one in Levy and Nikoukhah (2013), Xu et al. (2025). Such difference is also reflected in the resulting estimators: in U-RKF the robustification is applied to $P_{t|t}$, while in the estimators proposed in Levy and Nikoukhah (2013), Xu et al. (2025) the robustification is applied to P_t. From this viewpoint, the proposed estimator is more similar to the one in Abadeh et al. (2018) where the robustification is applied to $P_{t|t}$ and the ambiguity set is defined with respect to $\tilde{p}_t(w_t|Y_{t-1})$, i.e. the same conditional density that we have considered in the equivalent game, according to the Wasserstein distance. However, this estimator lacks an explicit interpretation of the underlying uncertainty in terms of (1) and (2), i.e. in terms of $p_t(x_{t+1}|x_t)$ and $\psi_t(y_t|x_t)$. In particular, it has not been investigated whether this uncertainty affects only the observation model (2) or not. Furthermore, it lacks an explicit characterization of the least favorable state space model, and does not provide any filter stability guarantee when the tolerance (i.e. Wasserstein radius) is chosen sufficiently small. Note that, in the limiting case where $c_t = 0$ (i.e. there is no model uncertainty), both U-RKF and the one in Levy and Nikoukhah (2013) coincide with the standard KF which represents the best estimator in the absence of uncertainty.

The least favorable model over the time interval $\{0,\dots,N\}$ is defined in (3), where $\tilde{\psi}_t = \tilde{\psi}_t^\star$ and $\tilde{\psi}_t^\star$ is the maximizer of (8). The next theorem provides a state-space representation approximating the least favorable model when the covariance matrix is taken sufficiently small in norm. The latter offers a tool

for generating worst-case data within the ambiguity set. Such data can be compared with actual measurements, thereby assessing the validity of the constructed ambiguity set and enabling a systematic evaluation of different filtering algorithms under the most challenging yet admissible conditions.

Theorem 4. *Let matrix Q be taken sufficiently small in norm, the least favorable model over the time horizon $t \in \{0 \dots N\}$ is approximated by:*

$$\begin{aligned} \eta_{t+1} &= \bar{A}_t \eta_t + \bar{B}_t \upsilon_t \\ y_t &= \bar{C}_t \eta_t + \bar{D}_t \upsilon_t . \end{aligned} \tag{28}$$

where $\eta_t := [\, x_t^\top \;\; e_{t-1}^\top \;\; \epsilon_{t-1}^\top \,]^\top \in \mathbb{R}^{3n}$; $\upsilon_t := [\, \epsilon_t^\top \;\; \upsilon_t^\top \,]^\top \in \mathbb{R}^{n+m}$ is white Gaussian noise with zero mean and covariance matrix $\Xi = \mathrm{diag}(Q, I_m)$,where $\mathrm{diag}(\cdot)$ denotes the block-diagonal matrix formed by placing its arguments along the main diagonal, and $\upsilon_t \in \mathbb{R}^m$ is normalized white Gaussian noise. Moreover,

$$\begin{aligned} \bar{A}_t &:= \begin{bmatrix} A & 0 & 0 \\ 0 & A - L_t CA - L_t F_t A & I_n - L_t F_t - L_t C \\ 0 & 0 & 0 \end{bmatrix} \\ \bar{B}_t &:= \begin{bmatrix} I_n & 0 \\ 0 & -L_t \Upsilon_t \\ I_n & 0 \end{bmatrix} \\ \bar{C}_t &:= \begin{bmatrix} C & F_t A & F_t \end{bmatrix}, \quad \bar{D}_t := \begin{bmatrix} 0 & \Upsilon_t \end{bmatrix}, \end{aligned} \tag{29}$$

where Υ_t is an arbitrary square root matrix of O_t, i.e. $\Upsilon_t \Upsilon_t^\top = O_t$, and

$$\begin{aligned} O_t &:= \left(R^{-1} - L_t^\top W_{t+1} L_t\right)^{-1} \\ F_t &:= -O_t L_t^\top W_{t+1} (I_n - L_t C). \end{aligned}$$

Finally,

$$W_{t+1} := \theta_t I_n + \Omega_{t+1}^{-1} \tag{30}$$

where Ω_t^{-1} is calculated by the following backward recursion:

$$\Omega_t^{-1} = A^\top F_t^\top O_t^{-1} F_t A + (A - L_t CA)^\top W_{t+1} (A - L_t CA) \tag{31}$$

with $\Omega_{N+1}^{-1} = 0$.

Proof. Let $e_t = x_t - \hat{x}_{t|t}$. Taking into account (1) and (2), we obtain

$$e_t = (A - L_t CA)\, e_{t-1} + (I_n - L_t C)\, \epsilon_{t-1} - L_t \varepsilon_t . \tag{32}$$

In view of (11), with $\theta_t = \lambda_t^{-1}$, we have the least favorable density is

$$\tilde{\psi}_t^\star(y_t|x_t) = \frac{1}{M_t} \exp\left(\frac{\theta_t}{2} \|e_t\|^2\right) \psi_t(y_t|x_t).$$

It is worth noting that the latter is not a normalized density, which implies that the hostile player has the opportunity to backtrack and change the least favorable density. Since only the model (2) is affected by uncertainty, the model for the state process in (1) does not change. Consider the nominal model for the observations (2), then it is not difficult to see that given x_t, there is a one-to-one correspondence between y_t and ε_t. Thus, we can characterize the least favorable model for the observations through ε_t. Notice that, ε_t does not depend on e_{t-1} and ϵ_{t-1} under the nominal model. Thus, the nominal density of ε_t is $\varphi_t(\varepsilon_t) \propto \exp(-\|\varepsilon_t\|_{R^{-1}}^2/2)$ where $\propto$ means that the two terms are the same up to constant scale factors. Instead, in view of (32), we make the guess that the least favorable density of ε_t is related to e_{t-1} and ϵ_{t-1}, namely, we consider $\tilde{\varphi}_t(\varepsilon_t|e_{t-1}, \epsilon_{t-1})$. Accordingly, we construct the term $\exp(\frac{1}{2}\|e_t\|_{\Omega_{t+1}^{-1}}^2)$ to indicate the cumulative error of the retroactive probability density changes of ε over the interval $[0, t]$. Here, Ω_t is a positive definite matrix of dimension n. Thus, the least favorable density of ε over the time interval $\{t+1 \dots N\}$ takes the form:

$$\begin{aligned} &\prod_{s=t+1}^{N} \exp\left(\frac{\theta_s}{2}\|e_s\|^2\right) \varphi_s(\varepsilon_s) \\ &\qquad \propto \exp\left(\frac{1}{2}\|e_t\|_{\Omega_{t+1}^{-1}}^2\right) \prod_{s=t+1}^{N} \tilde{\varphi}_s(\varepsilon_s|e_{s-1}, \epsilon_{s-1}) . \end{aligned} \tag{33}$$

Notice that, if the matrix Ω_t can be evaluated recursively, it is then possible to find the least favorable density $\tilde{\varphi}_t(\varepsilon_t|e_{t-1}, \epsilon_{t-1})$ through a backward recursion. Decreasing the time index t by 1 in (33) and subtracting it in (33), it is not difficult to see that

$$\begin{aligned} &\tilde{\varphi}_t(\varepsilon_t|e_{t-1}, \epsilon_{t-1}) \propto \\ &\quad \exp\left(-\frac{1}{2}\left(-\|e_t\|_{W_{t+1}}^2 + \|e_{t-1}\|_{\Omega_t^{-1}}^2 + \|\varepsilon_t\|_{R^{-1}}^2\right)\right), \end{aligned}$$

where W_{t+1} is defined in (30). Moreover, taking into account (32), we obtain the expression in (34) (see the equation in Box I).

Assuming that the covariance matrix Q is small in norm, ϵ_{t-1} is likely to take small values; therefore, the terms $\epsilon_{t-1}^\top (I_n - L_t C)^\top W_{t+1}(I_n - L_t C)\epsilon_{t-1}$ and $e_{t-1}^\top (A - L_t CA)^\top W_{t+1}(I_n - L_t C)\epsilon_{t-1}$ can be neglected. Under such approximation $\tilde{\varphi}_t$ can be expressed in the following compact way:

$$\begin{aligned} &\tilde{\varphi}_t(\varepsilon_t|e_{t-1}, \epsilon_{t-1}) \\ &\quad \propto \exp\left(-\frac{1}{2}\|\varepsilon_t - (F_t A_{t-1} e_{t-1} + F_t \epsilon_{t-1})\|_{O_t^{-1}}^2\right) \end{aligned} \tag{35}$$

where the backward recursion (31) is obtained by matching the quadratic term of e_{t-1} in (34) with the one in (35). In view of (35), we have

$$\tilde{\varphi}_t(\varepsilon_t|e_{t-1}, \epsilon_{t-1}) \sim \mathcal{N}(F_t A e_{t-1} + F_t \epsilon_{t-1}\,,\; O_t) .$$

Thus, the guess that ε_t depends on e_{t-1} and ϵ_{t-1} holds. We conclude that

$$\varepsilon_t = F_t A e_{t-1} + F_t \epsilon_{t-1} + \Upsilon_t \upsilon_t \tag{36}$$

where υ_t is normalized white Gaussian noise. Substituting (36) in (2) and (32), we obtain:

$$\begin{aligned} e_t &= (A - L_t CA)\, e_{t-1} + (I_n - L_t C)\, \epsilon_{t-1} \\ &\qquad - L_t(F_t A e_{t-1} + F_t \epsilon_{t-1} + \Upsilon_t \upsilon_t) \\ y_t &= C x_t + F_t A e_{t-1} + F_t \epsilon_{t-1} + \Upsilon_t \upsilon_t . \end{aligned}$$

Taking into account (1), $\eta_t := [\, x_t^\top \;\; e_{t-1}^\top \;\; \epsilon_{t-1}^\top \,]^\top$, $\upsilon_t := [\, \epsilon_t^\top \;\; \upsilon_t^\top \,]^\top$, and $\bar{A}_t$, $\bar{B}_t$, $\bar{C}_t$, $\bar{D}_t$ as in Eq. (29), then we obtain the following least favorable state space model:

$$\begin{aligned} \eta_{t+1} &= \bar{A}_t \eta_t + \bar{B}_t \upsilon_t \\ y_t &= \bar{C}_t \eta_t + \bar{D}_t \upsilon_t , \end{aligned}$$

which corresponds to the final form in (28). □

It is interesting to point out that the proposed ambiguity set (6) leads to a least favorable model whose structure is different from the one corresponding to (7), see Levy and Nikoukhah (2013, Section V). In particular, the former has state space dimension equal to $3n$, while the one in Levy and Nikoukhah (2013) has dimension $2n$. We conclude that, uncertainty only in the observation model leads to a least favorable model which is more complex than the one where uncertainty is in both the state process and observation models.

The theorem above also suggests the operative way to compute the least favorable model: first, we perform a forward recursion to compute L_t through Algorithm 1; then, we perform

$$\begin{aligned}\tilde{\varphi}_t(\varepsilon_t|e_{t-1},\epsilon_{t-1}) \propto \exp\Big(-\frac{1}{2}\big(&e_{t-1}^\top(\Omega_t^{-1}-(A-L_tCA)^\top W_{t+1}(A-L_tCA))e_{t-1}\\ &-\epsilon_{t-1}^\top(I_n-L_tC)^\top W_{t+1}(I_n-L_tC)\epsilon_{t-1}+\varepsilon_t^\top(R^{-1}-L_t^\top W_{t+1}L_t)\varepsilon_t\\ &+2e_{t-1}^\top(A-L_tCA)^\top W_{t+1}L_t\varepsilon_t+2\epsilon_{t-1}^\top(I_n-L_tC)^\top W_{t+1}L_t\varepsilon_t\\ &-2e_{t-1}^\top(A-L_tCA)^\top W_{t+1}(I_n-L_tC)\epsilon_{t-1}\big)\Big).\end{aligned} \tag{34}$$

Box I.

a backward recursion to compute Ω_t^{-1}; finally, we compute the state space matrices of (28).

We conclude this section by showing how to evaluate the performance of an arbitrary state estimator of the form

$$\hat{x}'_{t|t}=A\hat{x}'_{t-1|t-1}+L'_t(y_t-CA\hat{x}_{t-1|t-1}) \tag{37}$$

where L'_t is a Kalman gain sequence. Notice that, if we take $L'_t=L_t$, we obtain the U-RKF. Let $e'_t=x_t-\hat{x}'_{t|t}$ be the estimation error corresponding to (37), and let

$$\begin{aligned}&\Delta'_t=A-L'_tCA;\quad \Delta_t=A-L_tCA;\\ &\Lambda'_t=I_n-L'_tF_t-L'_tC;\quad \Lambda_t=I_n-L_tF_t-L_tC.\end{aligned}$$

Then, the covariance matrix of $\tilde{e}_t:=[\,(e'_t)^\top\ e_t^\top\ \epsilon_t^\top\,]^\top$ is obtained through the Lyapunov equation

$$\Pi_{t+1}=\Gamma_t\Pi_t\Gamma_t^\top+\mathrm{X}_t\Xi\mathrm{X}_t^\top \tag{38}$$

where

$$\Gamma_t=\begin{bmatrix}\Delta'_t & -L'_tF_tA & \Lambda'_t\\ 0 & \Delta_t-L_tF_tA & \Lambda_t\\ 0 & 0 & 0\end{bmatrix},\ \mathrm{X}_t=\begin{bmatrix}0 & -L'_t\Upsilon_t\\ 0 & -L_t\Upsilon_t\\ I_n & 0\end{bmatrix}.$$

Thus, the covariance matrix of the estimation error e'_t is given by the $n\times n$ submatrix of Π_t in the top-left position.

4. Filter stability

In this section we consider the situation in which the tolerance is constant, i.e. $c_t=c$. We assume that the pair (A,C) is observable. Notice that, the pair (A,Q) is reachable because $Q>0$. In what follows, we will show that it is possible to characterize an upper bound for the tolerance, say c_{MAX}, which guarantees that the gain L_t of U-RKF converges to a constant value for any $c\in(0,c_{MAX}]$. Then, we will show that for $c>0$ taken sufficiently small it is possible to guarantee in steady state that the estimation error under the least favorable model is bounded in mean square and the least favorable model is a state space model with constant parameters.

In view of Algorithm 1, P_t is characterized through the recursion

$$P_{t+1}=r_c(P_t):=A(P_t^{-1}+C^\top R^{-1}C-\theta_tI)^{-1}A^\top+Q \tag{39}$$

where we exploited the fact that the equation in Step 3 can be written as $P_{t|t}=(P_t^{-1}+C^\top R^{-1}C)^{-1}$. It is worth noting that the "distorted" Riccati operator r_c has the same structure of the one for the prediction-resilient Kalman filter (P-RKF) proposed in Levy and Nikoukhah (2013): the difference regards how θ_t depends on P_t. Such difference requires a convergence analysis which is substantially different to the one for the prediction-resilient case. Let $k\geq n$, we define the matrix

$$\mathbf{R}_k:=\mathcal{O}_k^\top(\mathcal{Q}_k+\mathcal{H}_k\mathcal{H}_k^\top)^{-1}\mathcal{O}_k+\mathcal{J}_k^\top S_k^{-1}\mathcal{J}_k$$

where

$$\begin{aligned}&S_k:=\mathcal{L}_k(I+\mathcal{H}_k^\top\mathcal{Q}_k^{-1}\mathcal{H}_k)^{-1}\mathcal{L}_k^\top-\phi_k^{-1}\otimes I\\ &\mathcal{J}_k:=\mathcal{O}_k^R-\mathcal{L}_k\mathcal{H}_k^\top[\mathcal{Q}_k+\mathcal{H}_k\mathcal{H}_k^\top]^{-1}\mathcal{O}_k\\ &\mathcal{O}_k:=[\,(CA^{N-1})^\top\ \dots\ (CA)^\top\ C^\top\,]^\top\\ &\mathcal{O}_k^R:=[\,(A^{N-1})^\top\ \dots\ A^\top\ I\,]^\top\\ &\mathcal{Q}_k:=I\otimes R\\ &\mathcal{H}_k:=\mathrm{Tp}\,(0\ H_1,H_2,\dots,H_{N-2},H_{N-1})\\ &\mathcal{L}_k:=\mathrm{Tp}\,(0,L_1,L_2,\dots,L_{N-2},L_{N-1})\\ &H_t:=CA^{t-1}Q^{1/2},\ L_t:=A^{t-1}Q^{1/2}\end{aligned}$$

where $\mathrm{Tp}(\cdot)$ denotes the block upper triangular Toeplitz matrix whose argument define its first block row and $Q^{1/2}$ denotes a square root matrix of Q. Matrix $\mathbf{R}_k$ is related to the observability Gramian of the k-fold composition of the mapping $r_c(\cdot)$, see Levy and Zorzi (2016, Section 4) for more details about mappings of this form. Let $\phi_k\in(0,\sigma_{max}(\mathcal{L}_k(I+\mathcal{H}_k^\top\mathcal{Q}_k^{-1}\mathcal{H}_k)^{-1}\mathcal{L}_k^\top))$ be the maximum value for which $\mathbf{R}_k$ is a positive definite matrix. As explained in Levy and Zorzi (2016), such ϕ_k does exist because (A,C) is observable. By Proposition 3.1 in Zorzi (2017), if

$$\theta_t\leq\phi_k,\ \forall t\geq q+1 \tag{40}$$

for some $q\in\mathbb{N}$, (A,Q) is reachable and (A,C) is observable, then the sequence generated by (39) converges and the corresponding algebraic equation admits a unique solution $P>0$. So, we have to find the condition on c for which (40) holds. In what follows, we consider the sequence generated by the Riccati operator

$$\begin{aligned}&\bar{P}_{t+1}=\bar{r}(\bar{P}_t):=A(\bar{P}_t^{-1}+C^\top R^{-1}C)^{-1}A^\top+Q,\\ &\bar{P}_0=Q\end{aligned}$$

and we define $\bar{P}_{t|t}=(\bar{P}_t^{-1}+C^\top R^{-1}C)^{-1}$.

Theorem 5. *Let model (1)–(2) be such that (A,C) is observable. We define*

$$c_{MAX}=\gamma(\bar{P}_{q|q},\phi_k)>0$$

where $k\geq n$ and $q\in\mathbb{N}$. Let c be such that $c\in(0,c_{MAX}]$, then the sequence generated by the iteration (39) converges to a unique solution $P>0$ for any $P_0>0$. Moreover, the limit L of the filtering gain L_t as $t\to\infty$ has the property that $A(I-LC)$ is stable.

Proof. By Lemma 4.1 in Zorzi (2017), we have that

$$P_t\geq\bar{P}_q,\quad t\geq q+1$$

for any $q\geq0$. Thus,

$$(P_t^{-1}+C^\top R^{-1}C)^{-1}\geq(\bar{P}_q^{-1}+C^\top R^{-1}C)^{-1}$$

which implies

$$P_{t|t}\geq\bar{P}_{q|q},\quad t\geq q+1.$$

Next, we prove that (40) holds: we prove that by contradiction, i.e. we assume that $\theta_t>\phi_k$ for some $t\geq q+1$. Since $\gamma(X,\cdot)$, with $X\geq0$ and $X\neq0$, is monotone increasing over $\mathbb{R}_+$ and

$\gamma(X,\theta) \geq \gamma(Y,\theta)$ for $X \geq Y$, see Lemma 4.3 in Zorzi (2017), it follows that

$$c = \gamma(P_{t|t},\theta_t) \geq \gamma(\bar{P}_{q|q},\theta_t) > \gamma(\bar{P}_{q|q},\phi_k) = c_{MAX}$$

which is a contradiction. Accordingly, all the hypotheses of Proposition 3.1 in Levy and Zorzi (2016) hold and thus (39) converges to a unique solution P. The stability of $A(I-LC)$ follows from the fact that the algebraic equation corresponding to (39) and having unique solution P, can be written as the Lyapunov equation

$$P = A(I-LC)P(I-LC)A^\top + Q + AL(AL)^\top. \quad \square$$

The least favorable model has been characterized over a simulation horizon $\{0\dots N\}$ through a forward and a backward recursion. Then, the least favorable model in steady state is obtained in the interval $t \in \{\lfloor \alpha N \rfloor \dots \lceil \beta N \rceil\}$ as $N \to \infty$ where α and β are such that $0 < \alpha < \beta < 1$. Let $t \in \{\lfloor \alpha N \rfloor \dots N\}$. By Theorem 5, $P_t \to P$, $\theta_t \to \theta$ and $L_t \to L$ as $N \to \infty$. So, the backward recursion (31) becomes

$$\Omega_t^{-1} = (F_tA)^\top O_t^{-1} F_t A + \bar{A}^\top W_{t+1} \bar{A} \tag{41}$$

with

$$O_t = (I - L^\top W_{t+1} L)^{-1}, \qquad F_t = -O_t L^\top W_{t+1}(I-LC)$$
$$\bar{A} = (I-LC)A, \qquad W_t = \Omega_t^{-1} + \theta I.$$

If Ω_t^{-1} (or equivalently W_t) converges as $N \to \infty$, then matrices $\bar{A}_t$, $\bar{B}_t$, $\bar{C}_t$, $\bar{D}_t$ converges and thus the least favorable model is a state space model with constant parameters in the steady state interval $t \in \{\lfloor \alpha N \rfloor \dots N\}$ as $N \to \infty$. Notice that, $\bar{A}$ and L depend on c through θ.

Proposition 6. *Assume that the map*

$$f : [0, \breve{\theta}] \to \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times m}$$
$$\theta \mapsto (\bar{A}, L)$$

is continuous for $\breve{\theta} > 0$ sufficiently small. Then, there exists $c > 0$ sufficiently small such that W_t, with $t \in \{0\dots\lceil \beta N \rceil\}$, converges to W as $N \to \infty$. Moreover, let $J = \bar{A}^\top WL(L^\top WL - I)^{-1}$, then $\bar{A} - LJ^\top$ is a stable matrix.

Proof. First, notice that (41) can be written as

$$\Omega_t^{-1} = \bar{A}^\top [W_{t+1} + W_{t+1}L(I - L^\top W_{t+1}L)^{-1}L^\top W_{t+1}]\bar{A}$$
$$= \bar{A}^\top (W_{t+1}^{-1} - LL^\top)^{-1}\bar{A}.$$

Adding on both sides θI we obtain

$$W_t = \bar{A}^\top (W_{t+1}^{-1} - LL^\top)^{-1}\bar{A} + \theta I$$

which is a Riccati recursion with terminal condition $W_N = \theta I$. The latter is similar to the one considered in Zorzi and Levy (2018) and using similar reasonings it is possible to prove that if $\theta > 0$ is sufficiently small (and thus f is continuous in a neighborhood of θ), then W_t converges and $\bar{A} - LJ^\top$ is a stable matrix. Notice that, P and θ are related through

$$c = \gamma((P^{-1} + C^\top R^{-1} C)^{-1}, \theta). \tag{42}$$

Moreover, P solves the algebraic form of the Riccati recursion (39), so $P \geq Q$ and thus

$$(P^{-1} + C^\top R^{-1} C)^{-1} \geq (Q^{-1} + C^\top R^{-1} C)^{-1} > 0. \tag{43}$$

Recall that $\gamma(X,\cdot)$, with $X \geq 0$ and $X \neq 0$, is monotone increasing over $\mathbb{R}_+$ and $\gamma(X,\theta) \geq \gamma(Y,\theta)$ for $X \geq Y$, Zorzi (2017, Lemma 4.3). Moreover, $\gamma(X,0) = 0$ for any $X \geq 0$ and the range of $[0, \sigma_{max}(X)^{-1})$ under $\gamma(X,\cdot)$ is $[0,\infty)$. Taking into account (42)–(43), we conclude that it is possible to take c sufficiently small such that θ is arbitrary small. $\square$

Let $e_t = x_t - \hat{x}_{t|t}$ be the state estimation error of U-RKF. The covariance matrix of e_t is given by the $n \times n$ top-left submatrix of Π_t which obeys the recursion in (38) with $L_t' = L_t$. In the case the least favorable model is in steady state, then the recursion takes the form as in (38). Since Γ_t and X_t depend on the constant parameters of the least favorable model and the filtering gain of U-RKF, we have $\Gamma_t \to \Gamma$, $X_t \to X$ as $t \to \infty$ and

$$\Gamma = \begin{bmatrix} (I-LC)A & -LFA & I_n - LF - LC \\ 0 & (I-LC)A - LFA & I_n - LF - LC \\ 0 & 0 & 0 \end{bmatrix}.$$

Matrix Γ is a stable matrix because its eigenvalues are the ones of $(I-LC)A$ and $(I-LC)A - LFA$ plus the ones in the origin. By Theorem 5, $A(I-LC)$ is a stable matrix for c sufficiently small. Since $A(I-LC)$ and $(I-LC)A$ have the same nonnull eigenvalues, we conclude that $(I-LC)A$ is stable. Moreover,

$$(I-LC)A - LFA = \bar{A} - JL^\top$$

which is a stable matrix for c sufficiently small by Proposition 6. Since Γ_t converges to a stable matrix, by Cattivelli and Sayed (2010, Theorem 1) we have that Π_t converges to the unique solution of the algebraic form of the Lyapunov recursion in (38). Accordingly, the covariance matrix of e_t is bounded in steady state.

5. Numerical examples

We present some numerical results to assess the proposed estimator. First, we analyze the performance of the filter in the worst case scenario. Then, we test it in a mass–spring-damper (MSD) system where uncertainties are primarily concentrated in the measurements.

5.1. Worst case analysis

We consider a nominal state space model of the form (1)–(2) with constant matrices

$$A = \begin{bmatrix} 0.95 & 1 \\ 0 & 1.2 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix},$$
$$C = \begin{bmatrix} 1 & 0 \\ 0.1 & -0.1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \tag{44}$$

and the initial state is modeled as a Gaussian random vector with zero mean and covariance matrix $P_0 = I$. Since such model is both reachable and observable, we compute the upper bound c_{MAX} for which we know that the proposed U-RKF converges. Given the matrices in (44), setting $k = 10$ and $q = 20$, then we found ϕ_k is approximately equal to 0.837, as shown in Fig. 1, and

$$\bar{P}_{q|q} = \begin{bmatrix} 0.4609 & 0.1727 \\ 0.1727 & 0.1036 \end{bmatrix}.$$

Thus, by Theorem 5, we have that $c_{MAX} = 0.5001$.

Next, we compare the performance of these estimators when applied to the least favorable model derived in Section 3. The constant tolerance, such that $c \leq c_{MAX}$, is the same for U-RKF, P-RKF and the least favorable model. Figs. 2 and 3 show the variance of the estimation error of the estimators, computed through (38) for $c = 0.1$ and $c = 5 \cdot 10^{-2}$. As we can see, the error variance converges for all the estimators. Also, the proposed U-RKF outperforms both P-RKF and KF. Moreover, two interesting and relevant aspects emerge. First, the larger the tolerance c is, the more evident the difference in performance becomes. Second, P-RKF outperforms KF. In plain words, P-RKF accounts for model uncertainties, but it is overly risk adverse in the sense that it considers model uncertainties also in the process Eq. (1). Finally,

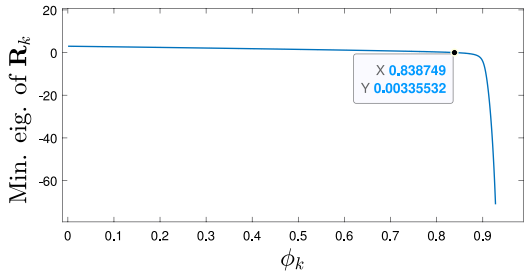


Fig. 1. Minimum eigenvalue of $\mathbf{R}_k$ as a function of ϕ_k with $k = 10$. The largest value of ϕ_k such that $\mathbf{R}_k$ is positive definite is approximately equal to 0.839.

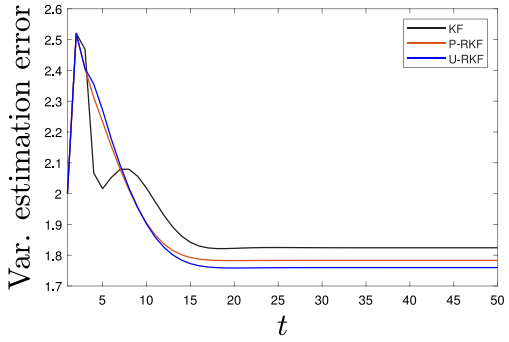


Fig. 2. Variance of the estimation error when KF (black line), P-RKF (red line) and U-RKF (blue line) are applied to the least favorable model with $c = 0.1$.

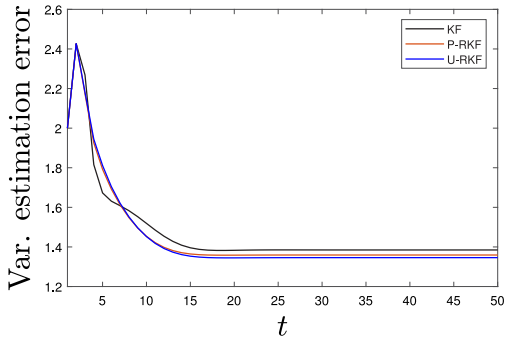


Fig. 3. Variance of the estimation error when KF (black line), P-RKF (red line) and U-RKF (blue line) are applied to the least favorable model with $c = 5 \cdot 10^{-2}$.

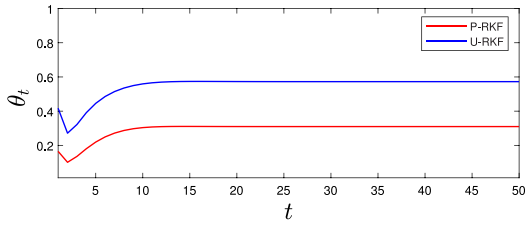


Fig. 4. Risk sensitivity parameter θ_t of P-RKF (red line) and U-RKF (blue line) when $c = 0.1$.

Fig. 4 illustrates the risk sensitivity parameter for U-RKF and P-RKF with $c = 0.1$. As we can see, although their Riccati recursions have the same structure, their risk sensitivity parameters are different. Finally, we also considered the situation in which the least favorable model is obtained by introducing small perturbations in the matrices A and Q (i.e. in the state process model). We found that U-RKF still outperforms both P-RKF and the KF.

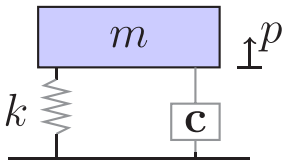


Fig. 5. Mass-spring-damper system.

5.2. MSD under sensor uncertainties

We consider a mass–spring-damper system, as shown in Fig. 5. The equation of the motion for this system is given by:

$$m\ddot{p} + \mathbf{c}(\dot{p} + \nu) + kp = F \tag{45}$$

where p represents the displacement of the object with mass $m = 0.1$ (kg) away from its resting position, $k = 5$ (N/m) is the spring constant, $\mathbf{c} = 2$ (Ns/m) is the damping coefficient and F denotes the external force, which is white Gaussian noise with zero mean and variance equal to 0.9; ν is white Gaussian noise with zero mean and variance equal to 0.09 which corresponds to the presence of a small "disturbance" force acting on the damper (e.g. the force generated by road irregularities, such as bumps and surface roughness, in a car suspension system). The displacement p is measured using a sensor with sampling time $T_s = 0.1s$. Let y_t denote the sensor measurement at time t, we consider the following four classical types of sensor uncertainties:

- Sensor drift $y_t = p_t + \tilde{\varepsilon}_t,\ \tilde{\varepsilon}_t \sim \mathcal{N}(0.1, R)$;
- Uniform noise $y_t = p_t + \tilde{\varepsilon}_t,\ \tilde{\varepsilon}_t \sim \mathcal{U}(-0.9, 1.1)$;
- Nonlinearity (dead zone)

$$y_t = \begin{cases} p_t + \tilde{\varepsilon}_t & \text{when } |p_t + \tilde{\varepsilon}_t| \geq 0.1 \\ 0 & \text{when } |p_t + \tilde{\varepsilon}_t| < 0.1 \end{cases}$$

  where $\tilde{\varepsilon}_t \sim \mathcal{N}(0, R)$;
- Outlier-contaminated noise

$$y_t = p_t + \tilde{\varepsilon}_t,\ \tilde{\varepsilon}_t \sim \begin{cases} \mathcal{N}(0, R) & \text{w.p.} \quad 0.9 \\ \mathcal{N}(0, 5R) & \text{w.p.} \quad 0.1 \end{cases}$$

  where w.p. stands for "with probability".

In the cases above $R = 0.25$. For each type of sensor uncertainty, we generate $M = 1000$ state and measurement trajectories with $N = 200$, i.e. the total time is 20s.

Our aim is to estimate the displacement using the sensors measurements previously generated. In doing that, we consider the state space model of (45) with state $x = [\,p \quad \dot{p}\,]^\top$ where we neglect the presence of the small disturbance force ν. Then, we discretize it with sampling time T_s obtaining an equation of the form (1). Finally, we assume that the nominal force is white Gaussian noise with zero mean and unit variance, i.e. its variance is slightly different than the actual one. In plain words, there is a mild mismatch between the nominal and the actual state process model. The nominal observation model is equal to (2) with $C = [\,1 \quad 0\,]^\top$ and ε_t is white Gaussian noise with zero and variance R. Then, we set $\hat{x}_0 = [\,0 \quad 0\,]^\top$ and $P_0 = 0.05I$.

We consider the following estimators which are based on the aforementioned nominal state-space model:

- KF denotes the standard Kalman filter;
- U-RKF denotes the proposed U-RKF with a fixed tolerance $c = 0.5$;
- P-RKF+or denotes the prediction-resilient Kalman filter proposed in Levy and Nikoukhah (2013); its tolerance is chosen in each realization through an *oracle*; the latter has access to the true state trajectory and chooses the tolerance minimizing the mean squared filtering error.

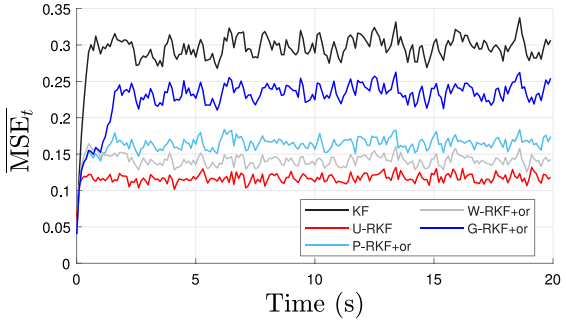


Fig. 6. Average variance of the displacement error for the different filters in the presence of sensor drift.

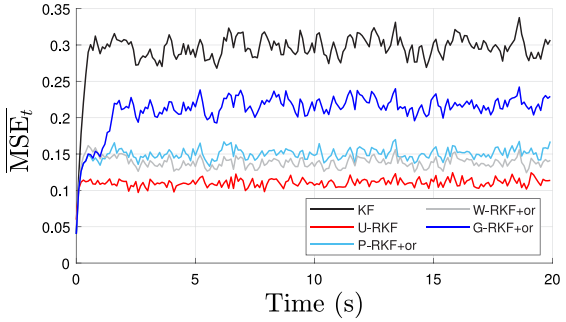


Fig. 8. Average variance of the displacement error for the different filters in the presence of nonlinearities.

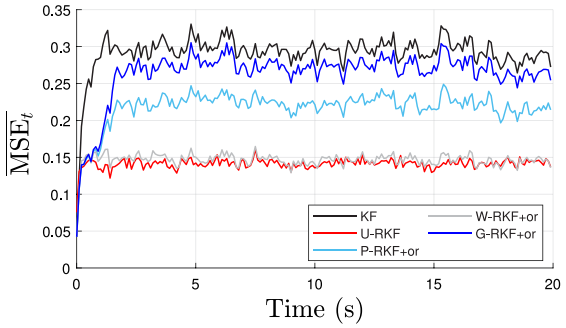


Fig. 7. Average variance of the displacement error for the different filters in the case of uniform noise.

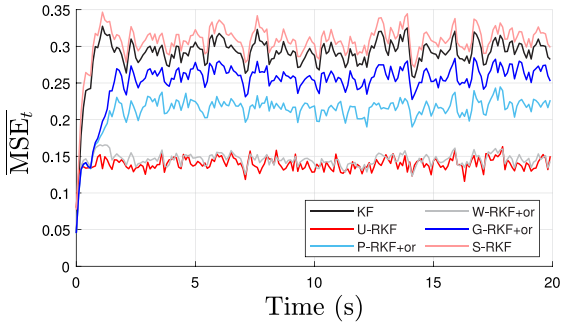


Fig. 9. Average variance of the displacement error for the different filters in the case of outlier-contaminated noise.

Table 1
Mean and standard deviation (Std) of the running time of U-RKF and W-RKF.

		Sensor drift	Uniform noise	Outlier-cont. noise	Nonlin.
U-RKF	Mean	0.0171s	0.0172s	0.0219s	0.0171s
	Std	0.0021s	0.0004s	0.0034s	0.0006s
W-RKF	Mean	4.4711s	4.4082s	4.5328s	4.4961s
	Std	0.0297s	0.0209s	0.5910s	0.1736s

- W-RKF+or denotes the Wasserstein distributionally robust filter proposed in Abadeh et al. (2018) where the tolerance[2] is chosen in each realization by an oracle whose definition is the same as before;
- G-RKF+or denotes the globalized robust filter proposed in Xu et al. (2025) whose parameter called "targeted level of error" is chosen in each realization by an oracle whose definition is the same as before;
- S-RKF denotes the sliding window variational outlier-robust filter proposed in Zhu et al. (2022) with the parameter settings suggested as in Table 4 of Zhu et al. (2022).

The oracles used in P-RKF+or, W-RKF+or and G-RKF+or search the optimal parameter (e.g. tolerance or targeted level of error which is constant over the time horizon) over a discretized interval of ten points whose extremal points are chosen in such a way the mean squared filtering error is properly captured without being overly broad, ensuring accuracy.

Then, we evaluate their performance through the average variance of the displacement error: $\overline{\text{MSE}}_t = \frac{1}{M}\sum_{k=1}^{M} \|\hat{p}_{t|t}^k - p_t^k\|^2$ where p_t^k and $\hat{p}_{t|t}^k$ denote the true displacement and the estimated one corresponding to the kth trajectory. Figs. 6–8 show the average variance of the displacement error for the estimators across different scenarios. It is worth stressing that the ambiguity set for the proposed U-RKF is fixed a priori, whereas for P-RKF, W-RKF and G-RKF, it is provided by the oracle. Even so, as we can see, U-RKF achieves the best performance across all three scenarios. The empirical evidence indicates that even when a mild uncertainty is present in the state process, suggesting that one might naturally consider ambiguity sets like those used in P-RKF+or and G-RKF+or, it is ultimately more effective to adopt U-RKF, as it better balances robustness and performance. In plain words, P-RKF+or and G-RKF+or are not able to split suitably the mismodeling budget between the state process and measurement models in this situation. As a consequence, these estimators are overly conservative in respect to the state process model and overly optimistic in respect to the observation model. Furthermore, W-RKF+or achieves a similar estimation performance to U-RKF because the uncertainty in both is framed in terms of the conditional density of w_t given Y_{t-1} as in (24). It is worth stressing that we did not report the performance of S-RKF in these cases because the comparison would not be fair, as S-RKF is specifically designed to handle the presence of outliers. Indeed, S-RKF performed worse than the robust estimators in all these cases. The performance of the estimators in the presence of outlier-contaminated noise is depicted in Fig. 9. As shown, U-RKF outperforms P-RKF+or and G-RKF+or, and it performs similarly to W-RKF+or. Moreover, its performance is also better than that of S-RKF. This is because S-RKF is more competitive in scenarios where outliers also affect the state process model (1).

To further assess how the choice of the tolerance c influences the performance of the proposed robust estimator, we conducted the following experiment. Specifically, we consider a range of

[2] In Abadeh et al. (2018) the tolerance parameter is called Wasserstein radius.

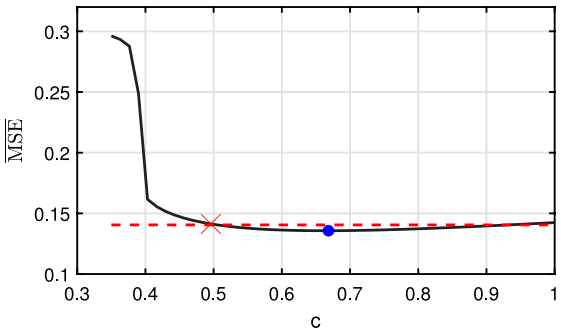


Fig. 10. $\overline{\text{MSE}}$ for U-RKF with $c \in \mathcal{C}$ (solid black line), W-RKF+or (red dashed line). The red cross denotes the $\overline{\text{MSE}}$ for $c = 0.5$ and the blue point the smallest value of $\overline{\text{MSE}}$.

values for c, that is $\mathcal{C} = [0.35,\ 1]$, and assess the performance of U-RKF, evaluating the average variance of the displacement error on the entire time horizon $\overline{\text{MSE}} = \frac{1}{N}\sum_{t=1}^{N} \overline{\text{MSE}}_t$. Fig. 10 reports the $\overline{\text{MSE}}$ of U-RKF as a function of $c \in \mathcal{C}$ in the case of outlier-contaminated noise (black solid line). The red cross indicates the $\overline{\text{MSE}}$ for U-RKF with $c = 0.5$ (i.e. the value considered in the previous analysis), while the blue point corresponds to the smallest value for the $\overline{\text{MSE}}$. As shown, the performance of U-RKF does not significantly change for c belonging to the interval $[0.5, 0.8]$. Finally, similar results were observed for the other three cases.

Finally, we compare the computational time of U-RKF and W-RKF, i.e. the estimators showing the best performance in respect to the others. Here, all simulations have been implemented in MATLAB and executed on a Mechanical Revolution notebook equipped with an AMD R9-7845HX CPU and a GeForce RTX 4070 GPU. To ensure a fair comparison, we also apply a fixed tolerance for W-RKF, set equal to the mean of the tolerances provided by the oracle over the $M = 1000$ realizations. In plain words, we consider a fair situation in which both algorithms compute the state estimate by considering only one tolerance. As shown in Table 1, the average running time per trial for U-RKF is approximately 0.02 s, while for W-RKF, it is around 4.5 s. Moreover, the standard deviations (Std) of the running times for both U-RKF and W-RKF are significantly smaller than their respective means, indicating high stability and a low coefficient of variation. This demonstrates that the proposed U-RKF is significantly more computationally efficient than W-RKF. The main reason is that W-RKF computes, at each time step, the filtering gain using a gradient-like method, which is computationally expensive. In contrast, in our algorithm it is only required to compute a scalar quantity, i.e. θ_t, which can be done using a bisection method, as done in Zenere and Zorzi (2018).

6. Update risk sensitive filter

The minimax problem (8) can be relaxed. More precisely, the constraint on the maximizer can be replaced by a penalty term in the objective function as follows:

$$\arg\min_{g_t \in \mathcal{G}_t} \max_{\tilde{\psi}_t \in \bar{\mathcal{B}}_t} J_t(\tilde{\psi}_t, g_t) - \theta \mathcal{D}(\tilde{\psi}_t, \psi_t) \tag{46}$$

where $\bar{\mathcal{B}}_t$ is the set of conditional densities of y_t given x_t and $\theta > 0$ is the risk sensitivity parameter which is a priori fixed. Thus, there is no need to determine θ at every time step, thereby reducing the computational complexity.

Algorithm 2 U-RSF at time step t

Input $\hat{x}_t$, P_t, θ, y_t, A, Q, C, R
Output $\hat{x}_{t|t}$, $\hat{x}_{t+1}$

$L_t = P_t C^\top (C P_t C^\top + R)^{-1}$
$\hat{x}_{t|t} = \hat{x}_t + L_t(y_t - C\hat{x}_t)$
$P_{t|t} = P_t - P_t C^\top (C P_t C^\top + R)^{-1} C P_t$
$V_{t|t} = (P_{t|t}^{-1} - \theta I)^{-1}$
$\hat{x}_{t+1} = A\hat{x}_{t|t}$
$P_{t+1} = A V_{t|t} A^\top + Q$

Theorem 7. *Consider the estimation problem corresponding to the state space model (1)–(2) and whose update estimate is obtained though (46). Assume that the initial state is Gaussian distributed as in (14). Then, the estimate of x_t given Y_t is obtained by (15)–(16) and the corresponding error covariance matrix is*

$$V_{t|t} = (P_{t|t}^{-1} - \theta I)^{-1}.$$

where $P_{t|t}$ is defined as in (18), and θ must be such that $0 < \theta < \sigma_{max}(P_{t|t})$. Moreover, condition (20) holds and the predictor takes the form (21)–(22).

Proof. Following the same reasonings in Lemma 2 and Theorem 3, it is not difficult to prove that the least favorable density takes the form in (11) where λ_t^{-1} is replaced by θ. Then, it is possible to show that the minimax problem (46) is equivalent to

$$\hat{x}_{t|t} = \arg\min_{g_t \in \mathcal{G}_t} \max_{\tilde{p}_t \in \check{\mathcal{B}}_t} \tilde{J}_t(\tilde{p}_t, g_t) - \theta \mathcal{D}(\tilde{p}_t, p_t),$$

where $\tilde{J}_t$ has been defined in (25), $\check{\mathcal{B}}_t$ is the set of conditional densities of $w_t := \begin{bmatrix} x_t^\top & y_t^\top \end{bmatrix}^\top$ given Y_{t-1}. Then, it is possible to show that $p_t(w_t|Y_{t-1})$ and its maximizer, i.e. $\tilde{p}_t^0(w_t|Y_{t-1})$, are both Gaussian. Thus, we can deduce that $\tilde{p}_t(x_{t+1}|Y_t)$ is Gaussian and the corresponding estimator. □

The resulting estimator, which will be called update risk sensitive filter (U-RSF), is outlined in Algorithm 2. Note that, U-RSF applies the distortion on $P_{t|t}$, which is different from the classic risk sensitive filter (Hassibi et al., 1999), hereafter called prediction risk sensitive filter (P-RSF). The latter applies the distortion on P_t because it assumes uncertainty is present in both (1) and (2). It remains to characterize the maximizer of (46), i.e. the least favorable model corresponding to U-RSF. Following similar reasonings as the ones in the proof of Theorem 4, it is possible to show that the least favorable model over the time interval $\{0 \dots N\}$ is the same to the one in (28), expect that θ_t in (30) is replaced by θ, i.e. $W_{t+1} := \theta I_n + \Omega_{t+1}^{-1}$.

6.1. Filter convergence

We consider the situation in which the pair (A, C) is observable. The pair (A, Q) is reachable because $Q > 0$. We show that it is possible to characterize an upper bound for the risk sensitivity parameter, say θ_{MAX}, which guarantees that the gain L_t is well defined and converges to a constant value as $t \to \infty$ for any $\theta \in (0, \theta_{MAX}]$. By Algorithm 2, the covariance matrix P_t obeys the recursion:

$$P_{t+1} = r_\theta^{RS}(P_t) := A(P_t^{-1} + C^\top R^{-1} C - \theta I)^{-1} A^\top + Q. \tag{47}$$

Under the reachability and observability assumptions, there exists $\phi_k > 0$, defined in the same way of the one in Section 4, such that if $\theta \le \phi_k$ then, by Proposition 3.1 in Levy and Zorzi (2016), the sequence generated by (47) converges and the corresponding algebraic equation admits a unique solution $P > 0$. However,

unlike the recursion (39), it is not guaranteed that $P_{t|t}^{-1} - \theta I$ is positive definite for any t. Next, we identify the conditions on P_0 which guarantee that $V_{t|t} > 0$ for any $t \geq 0$. It is not difficult to see that the recursion (47) can be written introducing an arbitrary observer gain matrix $G \in \mathbb{R}^{n\times m}$

$$r_\theta^{RS}(P) = (A-\alpha GC)(P^{-1} - \Psi_{\theta,\alpha})^{-1}(A-\alpha GC)^\top - X_{\theta,\alpha,P}\Phi_{\theta,\alpha,P}^{-1}X_{\theta,\alpha,P}^\top + GRG^\top + Q \tag{48}$$

where

$$\Psi_{\theta,\alpha} = (1-\alpha^2)C^\top R^{-1}C - \theta I$$
$$X_{\theta,\alpha,P} = \alpha(A-\alpha GC)(P^{-1} - \Psi_{\theta,\alpha})^{-1}C^\top - G$$
$$\Phi_{\theta,\alpha,P} = \alpha^2 C(P^{-1} - \Psi_{\theta,\alpha})^{-1}C^\top + R$$

and $0 < \alpha \leq 1$. Then, we consider the Lyapunov equation

$$\Sigma_{\rho,\alpha} = \rho^2(A-\alpha GC)\Sigma_{\rho,\alpha}(A-\alpha GC)^\top + GRG^\top + Q. \tag{49}$$

Since (A, C) is observable, we can choose α and G such that $A - \alpha GC$ is a stable matrix. Let $r < 1$ be the maximum among the modules of the eigenvalues of $A - \alpha GC$. Then, for $1 < \rho < r^{-1}$ matrix $\rho(A - \alpha GC)$ is a stable and the Lyapunov equation admits a unique solution. It is not difficult to see that the latter is also positive definite because (A, Q) is reachable. The next result shows, for α, G and ρ chosen as above, the conditions on P_0 which guarantees that $P_{t|t}^{-1} - \theta I$, or equivalently $V_{t|t}$, is positive definite.

Proposition 8. *Let*

$$\beta_{\rho,\alpha} = \sigma_{min}\left(\frac{\rho^2-1}{\rho^2}\Sigma_{\rho,\alpha}^{-1} + (1-\alpha^2)C^\top R^{-1}C\right).$$

If P_0 for the iteration (47) satisfies $0 < P_0 \leq \Sigma_{\rho,\alpha}$ and $0 \leq \theta \leq \beta_{\rho,\alpha}$, then $0 < P_t \leq \Sigma_{\rho,\alpha}$, and $V_{t|t} > 0$ for any $t \geq 0$.

Proof. First, we show that $P_t \leq \Sigma_{\rho,\alpha}$ implies that $V_{t|t} > 0$. Condition $\theta \leq \beta_{\rho,\alpha}$ is equivalent to

$$\frac{\rho^2-1}{\rho^2}\Sigma_{\rho,\alpha}^{-1} + (1-\alpha^2)C^\top R^{-1}C - \theta I \geq 0.$$

Since $\rho > 1$ and $0 < \alpha \leq 1$, it follows that $\Sigma_{\rho,\alpha}^{-1} + C^\top R^{-1}C - \theta I > 0$. Since $P_t \leq \Sigma_{\rho,\alpha}$, it follows that $P_t^{-1} + C^\top R^{-1}C - \theta I > 0$ and thus $V_{t|t} = (P_t^{-1} + C^\top R^{-1}C - \theta I)^{-1} > 0$. Next, we prove that $r^{RS}(\Sigma_{\rho,\alpha}) \leq \Sigma_{\rho,\alpha}$. Subtracting $r_\theta^{RS}(\Sigma_{\rho,\alpha})$ in (48) from (49) we have

$$\Sigma_{\rho,\alpha} - r_\theta^{RS}(\Sigma_{\rho,\alpha}) = X_{\theta,\alpha,P}\Phi_{\theta,\alpha,P}^{-1}X_{\theta,\alpha,P}^\top + (A-\alpha GC) \times (\rho^2\Sigma_{\rho,\alpha} - (\Sigma_{\rho,\alpha}^{-1} - \Psi_{\theta,\alpha})^{-1})(A-\alpha GC)^\top \geq 0$$

because condition $\theta \leq \beta_{\rho,\alpha}$ implies the conditions

$$\rho^2\Sigma_{\rho,\alpha} - (\Sigma_{\rho,\alpha}^{-1} - \Psi_{\theta,\alpha})^{-1} \geq 0, \quad \Phi_{\theta,\alpha,P} \geq 0.$$

Finally, we prove that if $P_0 \leq \Sigma_{\rho,\alpha}$, then $P_t \leq \Sigma_{\rho,\alpha}$. By the monotonicity of the operator r_θ^{RS}, see Levy and Zorzi (2016, Lemma 5.1), we have $P_1 = r_\theta^{RS}(P_0) \leq r_\theta^{RS}(\Sigma_{\rho,\alpha}) \leq \Sigma_{\rho,\alpha}$. By induction, assume that $P_t \leq \Sigma_{\rho,\alpha}$ then $P_{t+1} = r_\theta^{RS}(P_t) \leq r_\theta^{RS}(\Sigma_{\rho,\alpha}) \leq \Sigma_{\rho,\alpha}$. □

We are ready to establish the convergence result.

Theorem 9. *Let model (1)–(2) be such that (A, C) is observable. Let $0 < \alpha \leq 1$, $1 < \rho < r^{-1}$, $G \in \mathbb{R}^{n\times m}$, and ϕ_k with $k \geq n$, chosen as before. We define*

$$\theta_{MAX} = \min\{\beta_{\rho,\alpha}, \phi_k\} > 0.$$

If $\theta \in (0, \theta_{MAX}]$, then the sequence generated by the iteration (47) converges to a unique solution $P > 0$ for any $0 < P_0 \leq \Sigma_{\rho,\alpha}$. Moreover, the limit L of the filtering gain L_t as $t \to \infty$ has the property that $A(I - LC)$ is stable.

Proof. The convergence of L_t follows by the previous reasonings, i.e. the combination of Proposition 3.1 in Levy and Zorzi (2016) and Proposition 8. The stabilizing property of the limit of L_t can be proved likewise the proof of Theorem 5. □

In view of the above result, θ_{MAX} depends on ρ, α and G. Thus, it is possible to compute the best upper bound for θ optimizing θ_{MAX} with respect to (ρ, α, G). It is worth noting that (47) is the same recursion for the usual risk sensitive filter, i.e. P-RSF. However, there is a fundamental difference: while in Algorithm 2 we require that θ satisfies $P_t^{-1} + C^\top R^{-1}C - \theta I > 0$, in P-RSF we require the stronger condition $P_t^{-1} - \theta I > 0$. Thus, in general the range of θ in U-RSF is larger than the one for P-RSF. Moreover, the upper bound obtained through θ_{MAX} for the update case is in general larger than the prediction case. Indeed, forcing $\alpha = 1$ in Theorem 9 we obtain θ_{MAX} for the prediction case, see Levy and Zorzi (2016).

Regarding the least favorable model in steady state, we mention that it is possible to establish a result similar to Proposition 6 where the condition on c sufficiently small is replaced by θ sufficiently small. Then, using the same reasoning at the end of Section 4, it is possible to conclude that the covariance matrix of state estimation error of U-RSF in steady state is bounded.

6.2. Numerical examples

Consider the nominal state space model with the same constant matrices as those in (44) and the initial state is modeled as a Gaussian random vector with zero mean and covariance matrix $P_0 = 0.1I$. First, since such model is both reachable and observable, fixing $k = 10$, we found $\phi_k = 0.8387$. In view of Theorem 9, we know that θ_{MAX} depends on ρ, α, G. We maximize θ_{MAX} with $\alpha \in (0, 1]$, the entries of G lie in the interval $[-10, 10]$ and $\rho \in (1, (\sigma_{max}(A-\alpha GC))^{-1})$. We have found that the maximum value of θ_{MAX} for U-RSF is 0.5125. Moreover, the maximum value of θ_{MAX} for P-RSF, computed in the same way as for U-RSF, but with α fixed equal to 1, is 0.1709. We conclude that the upper bounds for θ_{MAX} of U-RSF and P-RSF are different and the one of U-RSF is larger than the one of P-RSF.

Next, we compare the performance of U-RSF, P-RSF, and KF applied to the least favorable model which is the maximizer of (46). We set $\theta = 0.17$ for U-RSF, P-RSF and the least favorable model. Fig. 11 shows that the proposed U-RSF performs better than P-RSF and the standard KF.

Finally, we assessed the performance of U-RSF using the numerical experiments from Section 5.2. The risk-sensitivity parameter θ was chosen in each realization through an oracle, which has access to the true state trajectory and selects the value that minimizes the mean squared filtering error. We found that U-RSF performs better than KF but worse than the other estimators. This inferior performance can be explained as follows: since U-RSF is a risk-sensitive-like filter, it concentrates the uncertainty at a single time instant when the model is most susceptible to model uncertainty, which is not very realistic.

7. Conclusion

In this paper, we have studied a robust state estimation problem where the uncertainty is primarily in the model for the observations. To address this, we have proposed a new robust estimation paradigm which is based on an ambiguity set that captures only the "mismatch" between the actual and nominal observation models. The resulting robust estimator exhibits a structure similar to that of the Kalman filter where the robustification takes place in the update stage. The latter is fundamentally different from the robust approaches in Levy and Nikoukhah (2013), Xu et al. (2025) where the robustification takes place in

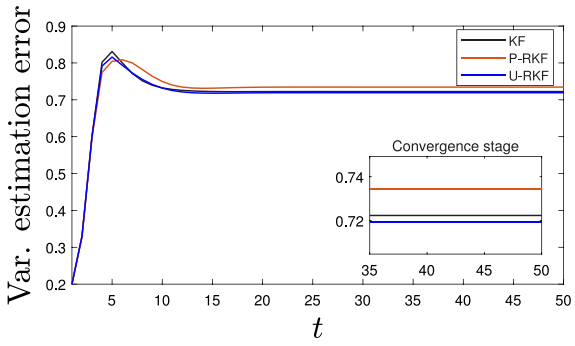


Fig. 11. Variance of the estimation error when the standard KF (black line), P-RSF (red line) and U-RSF (blue line) are applied to the least favorable model with $\theta = 0.17$.

the prediction stage. We have presented a numerical example based on a mass spring damper system, where sensor uncertainties constitute the dominant source of uncertainty. This example has shown that our estimator outperforms the ones in Levy and Nikoukhah (2013), Xu et al. (2025). Furthermore, this example has shown that the robust estimator proposed in Abadeh et al. (2018) (equipped with an oracle) performs similarly to our approach. Indeed, the robustification takes place in the update stage as in our estimator. Thus, our analysis seems to suggest that the estimator in Abadeh et al. (2018) postulated uncertainty only in the observation model. Finally, the numerical results showed that our approach is preferable since it is significantly more computationally efficient than the one in Abadeh et al. (2018).

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